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Introduction to Artificial Intelligence

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Eagle Delivery

This project explores how a simulated delivery robot can navigate between buildings on an Eastern Michigan University campus map. It compares A* search with a provided BFS baseline and uses Q-learning to learn a delivery policy through interaction with the environment. A custom map and controlled experiment are also used to investigate how a selected factor affects learning and performance.

The available actions are moving up, down, left, or right by one cell. A move to a traversable neighboring cell changes the robot's location; walls and map boundaries prevent movement through blocked cells. The goal test checks whether the robot has reached the selected destination. Each traversable move has a cost of 1, so the total path cost is the number of moves in the route.

Part A

Manhattan Distance and Admissibility: Manhattan distance is admissible in this environment because it never overestimates the minimum number of moves required to reach the destination. The robot must cover at least the difference in rows plus the difference in columns because diagonal movement is not allowed. Each move costs 1, so this distance is a lower bound on the

remaining path cost. Walls can force the robot to take a longer route, but they cannot produce a route shorter than the Manhattan distance.

A Search Comparison:*

Map	**Delivery Task**	**BFS Route Length**	**A* Route Length**	**BFS Expanded Node Count**	**A* Expanded Node Count**
Baseline	Student Center to Walton Hall	18	18	67	22
Baseline	Halle Library to Downing Hall	22	22	79	62
Baseline	Pray-Harrold to Putnam Hall	16	16	41	16
Detour	Student Center to Walton Hall	26	26	70	51
Detour	Halle Library to Downing Hall	22	22	65	60
Detour	Pray-Harrold to Putnam Hall	16	16	41	16

BFS and A* found routes of equal length for all six map and delivery combinations. This is expected because every move has the same cost and the Manhattan heuristic does not overestimate the remaining route cost. In these tests, A* expanded fewer nodes than BFS for

each delivery. The difference varied by task: for example, A* expanded 45 fewer nodes for Baseline S -> W, but only 5 fewer for Detour L -> D. While A* expands fewer nodes than BFS for each case in these tests, A* is not guaranteed to expand fewer nodes in every possible case.

Part B

Q-learning was trained for 2,000 episodes per task using seeds 0, 1, and 2. The baseline settings were Learning rate = 0.3, discount = 0.95, Exploration = 0.2, and slip probability at 0.0, with a maximum of 200 actions per episode. Each trained agent was evaluated for 100 episodes with exploration disabled and Q-values held fixed.

Under these settings, Q-learning achieved a 100% evaluation success rate on all three delivery tasks across all tested seeds. Mean successful delivery lengths were 18 steps for S → W, 22 for L → D, and 16 for H → P. These matched the shortest-path lengths found after utilizing A-star and BFS algorithms, indicating that the learned policies produced optimal routes for these tasks. Evaluation route lengths were identical across the three seeds.

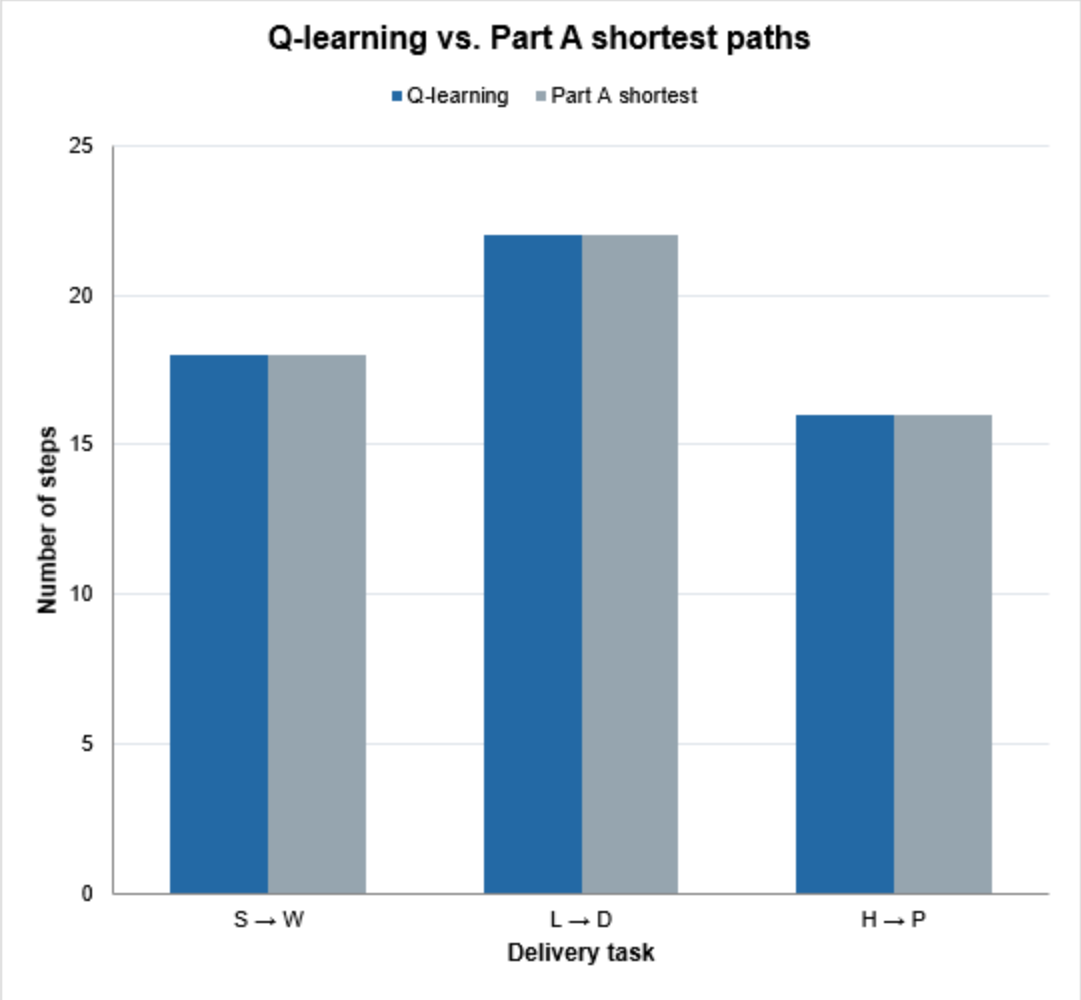


Figure 1*: Mean Q-learning evaluation delivery lengths compared with the Part A shortest-path lengths.*

Training performance improved rapidly, as shown in Figure 2. Mean returns in the first 100-episode block were -64.77 for S → W, -57.41 for L → D, and -23.59 for H → P. By the final block, these had increased to 17.42, 12.32, and 20.27, respectively. The improvement shows fewer failed deliveries and fewer actions per episode. S → W and L → D reached 100% training success in the second block, while H → P reached 100% in the third block.

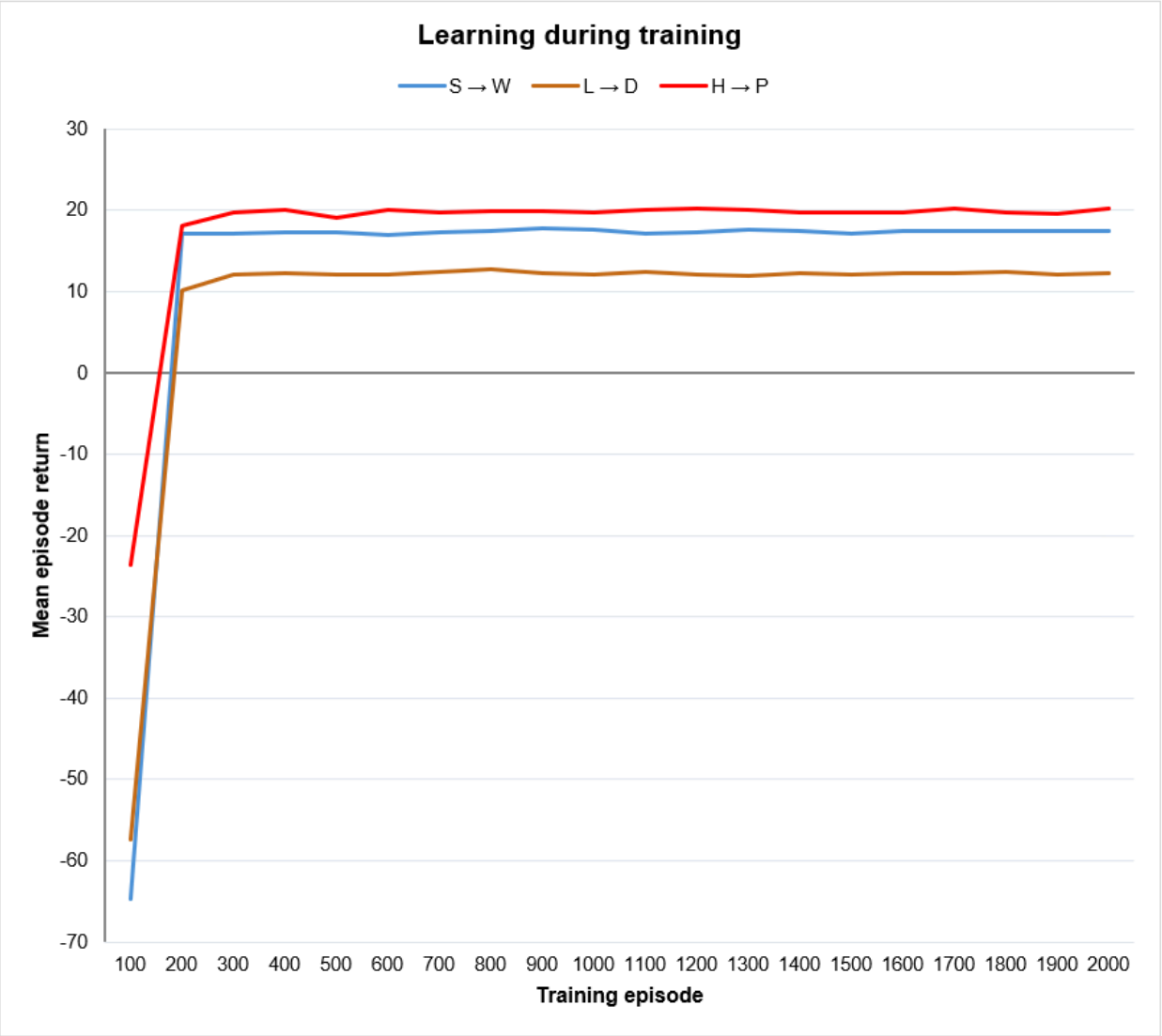


Figure 2*: Mean training return in non-overlapping 100-episode blocks, averaged across seeds 0, 1, and 2.*

Training returns remained below evaluation returns because exploration = 0.2 continued to introduce random actions during training. These actions could add unnecessary steps even after effective routes had been learned. During evaluation, exploration was disabled, producing mean returns of 22, 18, and 24 for S → W, L → D, and H → P. L → D had a lower evaluation

return because its shortest route required more steps. This does not indicate poorer learning, since all three tasks matched their respective shortest-path lengths.

Part C

Question: How does movement slip probability affect the learned policy's training progress and delivery performance on the custom map for the S -> W task?

Prediction: I predicted that increasing slip would increase the number of actions needed for a successful delivery because some attempted moves would fail. I also predicted that higher slip would make learning more difficult, reducing early training success. I expected success rates might decline at higher slip, especially with the 200-action episode limit.

Custom map: The custom map is 15 rows by 21 columns, with a blocked perimeter and all six building letters used once. There is also a dead-end branch at (row 2, column 3) that can only be accessed by the cell directly above it. Open corridors run horizontally across rows 1, 4, 7, 10, and 13, and vertical connectors join those corridors. For example, the corridors along row 1 and row 4, joined by the vertical passages at columns 1 and 5, form a loop around the blocked cells between them. For S -> W, the robot can travel down the left-side corridor and then across the bottom corridor, or travel across the upper corridor and descend along the right side. Both routes are 30 moves long. BFS and A* each found a 30-move route and expanded 143 nodes on this map. The loops provide alternative routes, while the dead-end is an unhelpful branch for reaching W.

Experiment: I used the custom map and the S -> W task for all conditions. I varied only the movement slip probability: 0.0, 0.1, and 0.2. The learning rate (alpha) was 0.3, the discount factor (gamma) was 0.95, and the exploration rate (epsilon) was held constant at 0.2 during

training. Each condition used seeds 0, 1, and 2, with a fresh Q-table for each seed. Each seed trained for 2,000 episodes, with a maximum of 200 actions per episode.

For evaluation, the agent was made to act greedily (epsilon = 0.0) for 100 episodes per seed. The same slip probability for that condition was used during both training and evaluation. Thus, each slip condition has 300 evaluation episodes across its three seeds. Success rate, mean successful delivery steps, and mean return were taken from the generated summary.csv files. The learning-curve CSVs report training success averaged across seeds in blocks of 100 episodes.

Findings: During episodes 1-100, mean training success was 63.7% at slip 0.0, 58.3% at slip 0.1, and 42.0% at slip 0.2. The slip 0.0 and 0.1 runs reached 100% in episodes 101-200. At slip 0.2, success was 99.0% in episodes 101-200 and reached 100% in episodes 201-300. Thus, higher slip reduced early training success and delayed perfect training success by one 100-episode block in the highest-slip condition.

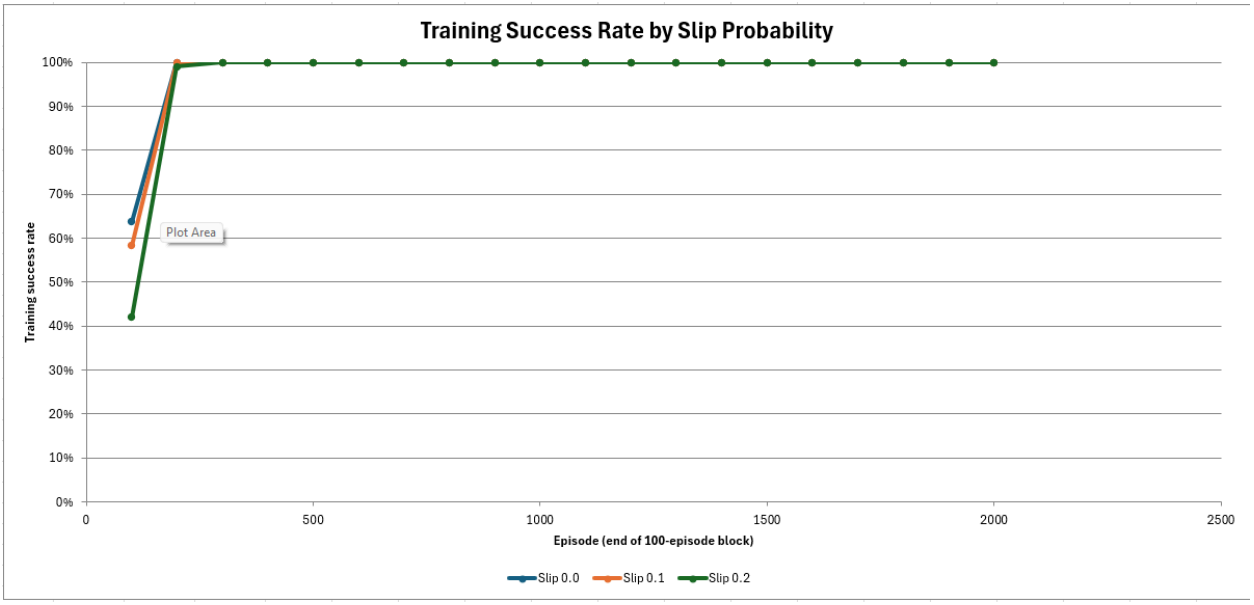


Figure 4: *Training success rate by episode end for slip probabilities 0.0, 0.1, and 0.2.*

Mean successful delivery length increased from 30.00 steps at slip 0.0 to 33.26 at slip 0.1 and 37.35 at slip 0.2. This supports the prediction that more movement failures require more actions to complete a delivery.

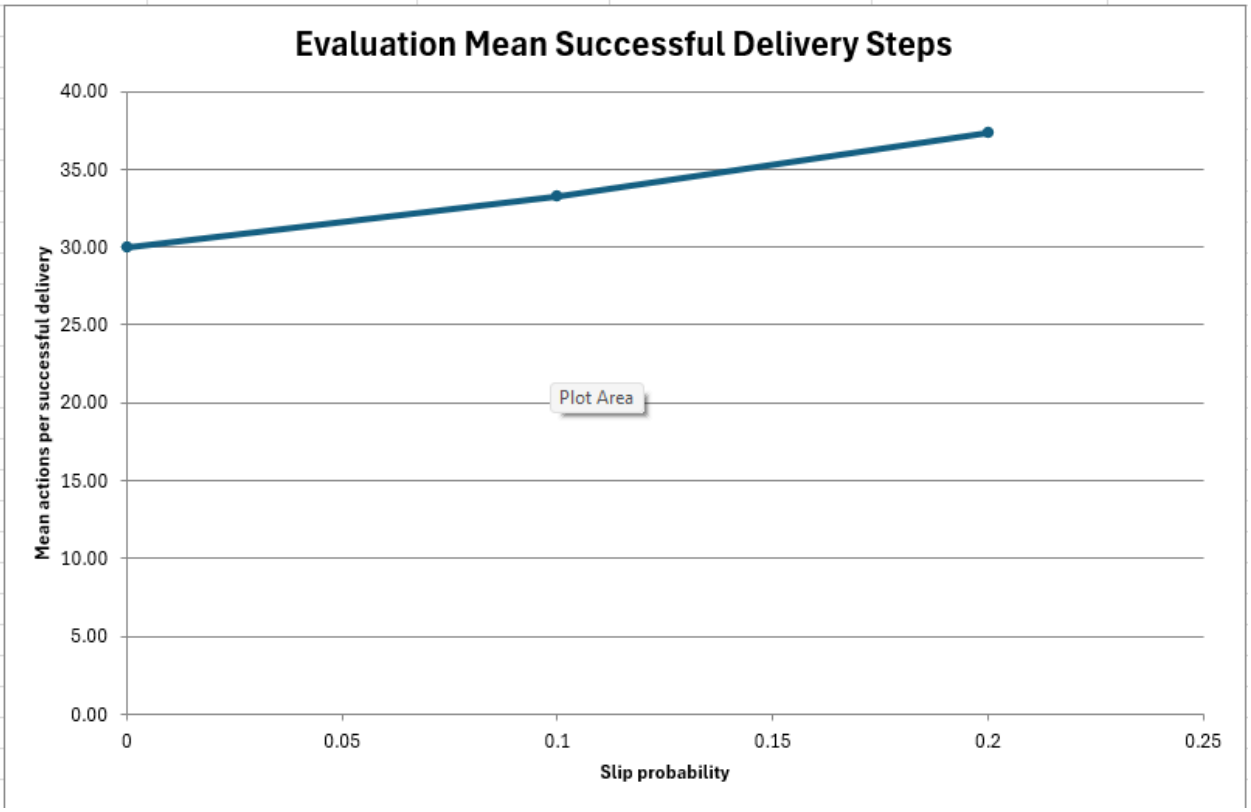


Figure 5: *Mean successful evaluation delivery steps by slip probability.*

Mean return declined from 10.00 at slip 0.0 to 6.74 at slip 0.1 and 2.65 at slip 0.2. This follows the increase in delivery length because a successful delivery of X actions has returned 40 - X.

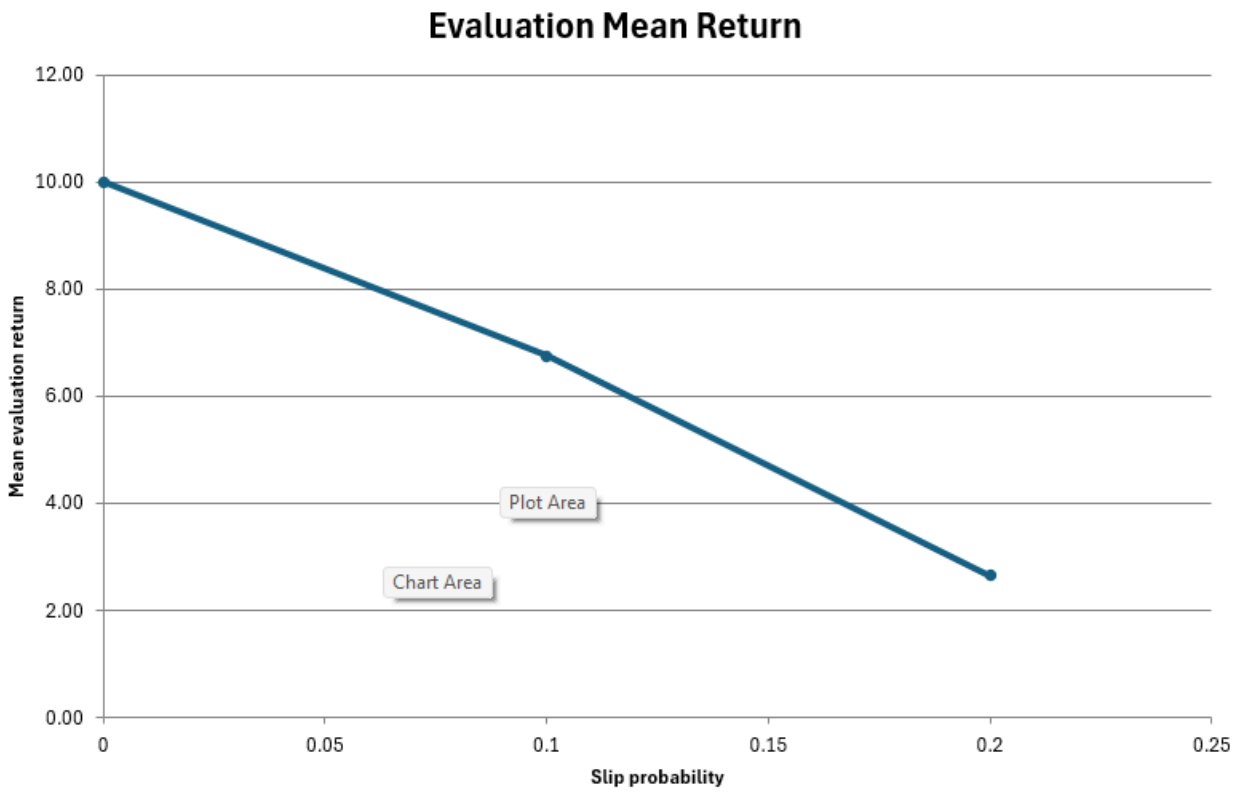


Figure 6: *Mean evaluation return by slip probability*

Conclusion: The first prediction was supported: increasing slip increased the mean number of actions in successful evaluations, from 30.00 with no slip to 33.26 at slip 0.1 and 37.35 at slip 0.2. These are averages across successful episodes and seeds. Since successful delivery return is 40 minus the number of actions, the mean return decreased accordingly, from 10.00 to 6.74 and 2.65.

The second prediction was also supported. In the first 100 training episodes, mean success fell from 63.7% with no slip to 58.3% at slip 0.1 and 42.0% at slip 0.2. At slip 0.2, success was 99.0% in episodes 101-200 and reached 100% in episodes 201-300, while both lower-slip conditions reached 100% in episodes 101-200. This suggests higher slip made early learning more difficult and delayed perfect training success by one 100-episode block.

The prediction that evaluation success might decline at higher slip, especially with the 200-action episode limit, was not observed in these runs. Evaluation success remained 100% (300 of 300 episodes) at all three slip settings, so no evaluation episode hit the action limit before delivering. Although slip made successful deliveries longer, their mean lengths remained well below 200 actions. Thus, for this map, task, and training setup, slip reduced early training success and delivery efficiency but did not reduce final evaluation success.

The findings are based on three independent seeds per condition and the specified 200-action limit.